TOPOLOGICAL CARTIER MODULES AND CYCLOTOMIC t-STRUCTURE

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ABSTRACT. These notes are for the talks at MCM homotopy reading seminar during 2026 fall. Our goal of these talks is to understand the cyclotomic t-structure introduced by [AN21], setting the stage for the subsequent talks regarding the Beilinson fiber square. We will also talk about various applications of this cyclotomic t-structure.

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INTRODUCTION

The goal of these talks is to understand the building blocks of cyclotomic spectra and the functor $\mathrm{TR}\colon \mathsf{CycSp} \to \mathsf{TCart}$. This plays a crucial role in the following paper [Ant+22]. We will essentially follow [AN21]. Our main theorem is:

Theorem A. *There exists an accessible, left complete t-structure on CycSp_p which is compatible with the symmetric monoidal structure. Such that:*

(1) The connective part $(\mathsf{CycSp}_p)_{\geq 0}$ is those cyclotomic spectra such that the underlying spectra is connective.
(2) The heart $\mathsf{CycSp}_p^{\heartsuit}$ is identified with the abelian category of (derived) V-complete Cartier modules.

The existence of the t-structure is quite formal, However, it demands efforts to identify the heart. The strategy to prove this theorem is as follows. We first define the topological counterpart of Cartier modules, which is the category of *topological Cartier modules* denoted by TCart. The category TCart is equipped with a t-structure and an adjunction

$$(-)/V\colon \mathsf{TCart} \rightleftharpoons \mathsf{CycSp}_p\colon \mathrm{TR}(-)$$

The right adjoint $\mathrm{TR}(-)$ is fully faithful and exact, with essential image identified with the V-complete topological Cartier modules on bounded below objects.

1. Cyclotomic t-structure

We recollect the definition and basic properties of cyclotomic spectra from [NS18]. We will work with p-typical cyclotomic spectra most time.

Definition 1.1. The ∞-category of p-typical cyclotomic spectra is defined as

$$\mathsf{CycSp}_p := \mathrm{LEq}\Big(\mathsf{Sp}^{BS^1} \underset{(-)^{tC_p}}{\overset{\mathrm{id}}{\rightrightarrows}} \mathsf{Sp}^{BS^1}\Big)$$

This can be written as

$$\begin{array}{ccc} \mathsf{CycSp}_p & \longrightarrow & \mathrm{Fun}(\Delta^1, \mathsf{Sp}^{BS^1}) \\ \downarrow & & \downarrow{\scriptstyle (\mathrm{ev}_0, \mathrm{ev}_1)} \\ \mathsf{Sp}^{BS^1} & \xrightarrow[(\mathrm{id}, (-)^{tC_p})]{} & \mathsf{Sp}^{BS^1} \times \mathsf{Sp}^{BS^1} \end{array}$$

Hence a p-typical cyclotomic spectrum is a spectrum X equipped with a S^1-action and a S^1-equivariant map $\varphi\colon X \to X^{tC_p}$.

We let $(\mathsf{CycSp}_p)_{\geq 0}$ denote the full subcategory of CycSp_p generated by those cyclotomic spectra with connective underlying spectra. It is clear that the unit $\mathbb{S}^{\mathrm{triv}}$ sits in $(\mathsf{CycSp}_p)_{\geq 0}$.

Lemma 1.2. *The natural transformation $\tau_{\geq 0}(-)^{tC_p} \to (-)^{tC_p}$ induces an equivalence*

$$(\mathsf{CycSp}_p)_{\geq 0} \simeq \mathrm{LEq}(\mathrm{id}, \tau_{\geq 0}(-)^{tC_p} \colon \mathsf{Sp}_{\geq 0}^{BS^1} \rightrightarrows \mathsf{Sp}_{\geq 0}^{BS^1})$$

As a result, $(\mathsf{CycSp}_p)_{\geq 0}$ is presentable and the inclusion to CycSp_p preserves colimits.

Proof. The equivalence is because any cyclotomic structure over a connective spectrum is determined by its factorization through the connective cover. The second claim follows from the fact that the forgetful functors $\mathsf{CycSp}_p \to \mathsf{Sp}^{BS^1}$ and $(\mathsf{CycSp}_p)_{\geq 0} \to \mathsf{Sp}_{\geq 0}^{BS^1}$ preserve and reflect colimits, together with the fact that the inclusion $\mathsf{Sp}_{\geq 0}^{BS^1} \to \mathsf{Sp}^{BS^1}$ preserves colimits. □

We are going to prove the first half of the main theorem, which largely relies on the following lemma of [Lur17, Proposition 1.4.4.11].

Lemma 1.3. *Let $\mathcal{C}$ be a stable ∞-category. Let $\mathcal{C}' \subset \mathcal{C}$ be a full subcategory which is presentable, closed under colimits and closed under extensions. Then there exists a t-structure over $\mathcal{C}$ such that $\mathcal{C}_{\geq 0} \simeq \mathcal{C}'$.*

Theorem 1.4. *The ∞-category $(\mathsf{CycSp}_p)_{\geq 0}$ forms the connective part of a t-structure over CycSp_p, which is left complete, accessible and compatible with the symmetric monoidal structure.*

Recall that a t-structure is said to be *accessible* if its connective part is presentable. We say a t-structure is compatible with symmetric monoidal structure if the connective part contains the unit and is closed under $\otimes$.

Proof. It is clear that $(\mathsf{CycSp}_p)_{\geq 0}$ is closed under extension. It follows immediately from Lemma 1.2 and Lemma 1.3 that it determines the connective part of a t-structure. It remains to show the left completeness. Note that by a same argument as in [NS18, Chapter I.2], we can show that $(\mathsf{CycSp}_p)_{\geq 0}$ is closed under small products. Therefore by [Lur17, Proposition 1.2.1.19], the left completeness is equivalent to left separateness. Which is saying that

$$\bigcap_{n\in\mathbb{Z}} (\mathsf{CycSp}_p)_{\geq n} \simeq 0$$

It is clear since the underlying spectrum of the objects in this intersection is zero. □

We conclude with some examples regarding the heart.

Example 1.5. If $X \in (\mathsf{CycSp}_p)_{\leq 0}$, then X must satisfy $\pi_i \mathrm{TC}(X) = 0$ for every $i > 0$ since $\mathrm{TC}(X) \simeq \mathrm{map}_{\mathrm{cyc}}(\mathbb{S}^{\mathrm{triv}}, X)$ and $\mathbb{S}^{\mathrm{triv}}$ lies in $(\mathsf{CycSp}_p)_{\geq 0}$. In particular, $\mathbb{Z}^{\mathrm{triv}}$ does not lie in $\mathsf{CycSp}_p^{\heartsuit}$ for $\pi_i \mathrm{TC}(\mathbb{Z}^{\mathrm{triv}}) \cong \mathbb{Z}_p$ for every positive odd i.

Example 1.6. We have a symmetric monoidal localization functor $\pi_0^{\mathrm{cyc}} \colon (\mathsf{CycSp}_p)_{\geq 0} \to \mathsf{CycSp}_p^{\heartsuit}$. Therefore, $\mathsf{CycSp}_p^{\heartsuit}$ inherits a closed symmetric monoidal structure.

In order to study closely the heart of the t-structure, we need to compare CycSp_p with another category, namely the category of (p-typical) topological Cartier modules.

2. Topological Cartier modules

Recall that classically, a p-typical Cartier module is an abelian group M equipped with two endomorphisms V and F, called the *Verschiebung* and *Frobenius*, such that $F \circ V = p$. We now introduce a topological generalization of this.

2.1. Topological Cartier modules.

Definition 2.1. A *p-typical topological Cartier module* is a spectrum M with S^1-action, together with an S^1-equivariant factorization of the C_p-norm

$$M_{hC_p} \xrightarrow{V} M \xrightarrow{F} M^{hC_p}$$

where M_{hC_p} and M^{hC_p} carries the residue $S^1 \simeq S^1/C_p$-action.

Example 2.2. For any topological Cartier module with trivial S^1-action, we essentially have a factorization

$$M \longrightarrow M_{hC_p} \xrightarrow{V} M \xrightarrow{F} M^{hC_p} \longrightarrow M$$

where the composition is multiplication by p. In particular, any classical Cartier module becomes a topological Cartier module.

Example 2.3. Let M be a topological Cartier module, then the above decomposition

$$M \longrightarrow M_{hC_p} \xrightarrow{V} M \qquad\qquad M \xrightarrow{F} M^{hC_p} \longrightarrow M$$

gives a classical Cartier module structure over $\pi_n(M)$ for every $n \in \mathbb{Z}$.

We now formalize the definition of $(\infty, 1)$-category of topological Cartier modules.

Definition 2.4. The ∞-category of p-typical topological Cartier modules TCart_p is defined as the pullback

$$\begin{array}{ccc} \mathsf{TCart}_p & \longrightarrow & \mathrm{Fun}(\Delta^2, \mathsf{Sp}^{BS^1}) \\ \downarrow & & \downarrow{\scriptstyle (\mathrm{ev}_1, \partial_1)} \\ \mathsf{Sp}^{BS^1} & \xrightarrow[(\mathrm{id}, \mathrm{Nm}_{C_p})]{} & \mathsf{Sp}^{BS^1} \times \mathrm{Fun}(\Delta^1, \mathsf{Sp}^{BS^1}) \end{array}$$

One can think of objects in TCart_p as pairs (M, V_M, F_M, σ_M), where M is a spectrum with S^1-action, V_M and F_M are decomposition maps as in Definition 2.1, and σ_M represents a 2-simplex that encodes $F \circ V \simeq \mathrm{Nm}_{C_p}$.

Proposition 2.5. *The ∞-category TCart_p is stable, presentable, and the forgetful functor $\mathsf{TCart}_p \to \mathsf{Sp}^{BS^1}$ preserves colimits.*

Proof. The ∞-category TCart_p is stable since it is the pullback of stable ∞-categories along exact functors. TCart_p is accessible since it is the pullback of accessible categories along accessible functors. It suffices to show that TCart_p is closed under colimits. For any diagram K, one can define the Verschiebung and Frobenius on $\mathrm{colim}_{k\in K} M(k)$ as the composition

$$\big(\mathrm{colim}_{k\in K} M(k)\big)_{hC_p} \simeq \mathrm{colim}_{k\in K} M(k)_{hC_p} \xrightarrow{\mathrm{colim}_{k\in K} V_{M(k)}} \mathrm{colim}_{k\in K} M(k)$$

and

$$\mathrm{colim}_{k\in K} M(k) \xrightarrow{\mathrm{colim}_{k\in K} F_{M(k)}} \mathrm{colim}_{k\in K} M(k)^{hC_p} \to \big(\mathrm{colim}_{k\in K} M(k)\big)^{hC_p}$$

These two indeed defines a factorization of Nm_{C_p} by functoriality of Nm_{C_p}:

$$\begin{array}{ccc} \mathrm{colim}_{k\in K} M(k)_{hC_p} & \xrightarrow{\simeq} & \big(\mathrm{colim}_{k\in K} M(k)\big)_{hC_p} \\ {\scriptstyle \mathrm{colim}_k \mathrm{Nm}_{C_p}}\downarrow & & \downarrow{\scriptstyle \mathrm{Nm}_{C_p}} \\ \mathrm{colim}_{k\in K} M(k)^{hC_p} & \longrightarrow & \big(\mathrm{colim}_{k\in K} M(k)\big)^{hC_p} \end{array}$$

Moreover, using the same argument one can shows the forget functor also preserves limits. □

Remark 2.6. We can formally shows that TCart_p is generated by a single compact generator. If we denote the left adjoint of the forgetful functor $\mathsf{TCart}_p \to \mathsf{Sp}$ by L, then TCart_p is compactly generated by $K = L(\mathbb{S})$. Indeed, K is compact since we have

$$\mathrm{Map}_{\mathsf{TCart}_p}(K, \mathrm{colim}_i M_i) \simeq \mathrm{Map}_{\mathsf{Sp}}(\mathbb{S}, \mathrm{colim}_i M_i) \simeq \mathrm{colim}_i \mathrm{Map}_{\mathsf{Sp}}(\mathbb{S}, M_i) \simeq \mathrm{colim}_i \mathrm{Map}_{\mathsf{TCart}_p}(K, M_i)$$

It is a generator since $\mathrm{Map}_{\mathsf{TCart}_p}(K, M) \simeq 0$ simply implies $M \simeq 0$ by adjunction.

Remark 2.7. Let $\mathsf{CycSp}_p^{\mathrm{Fr}}$ denote the category of p-typical cyclotomic spectra with Frobenius lift. That is, the category of S^1-equivariant spectra X with an S^1-equivariant map $\psi_p \colon X \to X^{hC_p}$. Then there is a sequence of forgetful functors

$$\mathsf{TCart}_p \xrightarrow{u_1} \mathsf{CycSp}_p^{\mathrm{Fr}} \xrightarrow{u_2} \mathsf{CycSp}_p$$

where u_2 is the obvious forgetful and u_1 sends $(M, V_M, F_M, \sigma_M) \in \mathsf{TCart}_p$ to (M, F_M). Moreover, one can show that we have a pullback diagram.

$$\begin{array}{ccc} \mathsf{TCart}_p & \xrightarrow{\theta} & \mathsf{Sp}^{BS^1} \times \mathsf{Sp}^{BS^1} \\ {\scriptstyle u_1}\downarrow & & \downarrow{\scriptstyle 0} \\ \mathsf{CycSp}_p^{\mathrm{Fr}} & \xrightarrow[\theta_0]{} & \mathrm{Fun}(\Delta^1, \mathsf{Sp}^{BS^1}) \end{array}$$

Here the upper horizontal map θ sends $(M, V_M, F_M, \sigma_M) \in \mathsf{TCart}_p$ to $(M_{hC_p}, M^{hC_p}/F)$ and the lower horizontal map sends (M, ψ_p) to the composition $M_{hC_p} \xrightarrow{\mathrm{Nm}_{C_p}} M^{hC_p} \to M^{hC_p}/F$. In fact, this follows from the following defining diagram where each square is a pullback square.

$$\begin{array}{ccccc} \mathsf{TCart}_p & \longrightarrow & \mathrm{Fun}(\Delta^2, \mathsf{Sp}^{BS^1}) & \xrightarrow{(\mathrm{ev}_0, \mathrm{cofib}(\partial_1))} & \mathsf{Sp}^{BS^1} \times \mathsf{Sp}^{BS^1} \\ {\scriptstyle u_1}\downarrow & & {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle 0} \\ \mathsf{CycSp}_p^{\mathrm{Fr}} & \longrightarrow & \mathrm{Fun}(\Lambda^2_2, \mathsf{Sp}^{BS^1}) & \xrightarrow{\partial_0} & \mathrm{Fun}(\Delta^1, \mathsf{Sp}^{BS^1}) \\ {\scriptstyle u_2}\downarrow & & {\scriptstyle (\mathrm{ev}_1, \partial_1)}\downarrow & & \downarrow{\scriptstyle (\mathrm{ev}_0, \mathrm{ev}_1)} \\ \mathsf{Sp}^{BS^1} & \xrightarrow{(\mathrm{id}, \mathrm{Nm}_{C_p})} & \mathsf{Sp}^{BS^1} \times \mathrm{Fun}(\Delta^1, \mathsf{Sp}^{BS^1}) & \xrightarrow{(\mathrm{id}, \mathrm{ev}_1)} & \mathsf{Sp}^{BS^1} \times \mathsf{Sp}^{BS^1} \end{array}$$

Lemma 2.8. *Suppose we are given two topological Cartier modules M and N, then there is a fiber sequence of mapping spectra:*

$$\mathrm{map}_{\mathsf{TCart}_p}(M, N) \longrightarrow \mathrm{map}_{\mathsf{CycSp}_p^{\mathrm{Fr}}}(M, N) \longrightarrow \mathrm{map}_{\mathsf{Sp}^{BS^1}}(M_{hC_p}, \mathrm{fib} F_N)$$

Proof. We can apply Remark 2.7 and identify the cofiber of the mapping spectrum. □

Remark 2.9. The forgetful functor $U\colon \mathsf{Sp}^{BS^1} \to \mathsf{Sp}$ is plethyistic, which gives a comonad $(-)^{hC_p}$ and a monad $(-)_{hC_p}$ over Sp^{BS^1}. It is clear that $\mathrm{CoAlg}_{(-)^{hC_p}} \simeq \mathsf{CycSp}_p^{\mathrm{Fr}}$. Dual to Lemma 2.8, we can also gives a cofiber sequence

$$\mathrm{map}_{\mathsf{TCart}_p}(M, N) \longrightarrow \mathrm{map}_{\mathrm{Alg}_{(-)_{hC_p}}}(M, N) \longrightarrow \mathrm{map}_{\mathsf{Sp}^{BS^1}}(\mathrm{cofib}(V_M), N^{hC_p})$$

We can define a t-structure over the ∞-category TCart_p as follows, this t-structure is relatively easy to capture, which will allow us to compare to the t-structure over the ∞-category of p-typical cyclotomic spectra.

Proposition 2.10. *There exists an accessible, bicomplete t-structure over TCart_p that satisfies the following properties:*

1. *The connective part $(\mathsf{TCart}_p)_{\geq 0}$ is identified with those topological Cartier modules whose underlying spectra are connective.*
2. *The coconnective part $(\mathsf{TCart}_p)_{\leq 0}$ is identified with those topological Cartier modules whose underlying spectra are coconnective.*
3. *The heart $\mathsf{TCart}_p^{\heartsuit}$ is identified with the abelian category of classical Cartier modules.*
4. *The forgetful functor $\mathsf{TCart}_p \to \mathsf{Sp}^{BS^1}$ is t-exact.*

Proof. The assertion (1) is formally derived from [Lur17, Proposition 1.4.4.11]. The remaining task is to identify the coconnective part, which we denote by $\mathcal{C}_{\leq 0}$. It follows from Lemma 2.8

that we always have $\mathrm{Map}_{\mathsf{TCart}_p}(\mathsf{TCart}_{\geq 0}, \mathsf{TCart}_{\leq -1}) \simeq 0$. It remains to show that for any $M \in \mathsf{TCart}_p$, regarding as a spectrum, $\tau_{\leq 0}M$ and $\tau_{\geq 0}$ inherit a topological Cartier module structure. It suffices to show this for $\tau_{\geq 0}M$ since the forgetful functor $\mathsf{TCart}_p \to \mathsf{Sp}$ is exact and we can take the cofiber. The Verschiebung and the Frobenius are exhibited by the factor through and by the fact that $\tau_{\geq 0}(-)$ commutes with limits.

$$\begin{array}{ccc} (\tau_{\geq 0}M)_{hC_p} & \xrightarrow{V'} & \tau_{\geq 0}M \\ \downarrow & & \downarrow \\ M_{hC_p} & \xrightarrow{V} & M \end{array} \qquad \begin{array}{ccc} \tau_{\geq 0}M & \xrightarrow{F'} & (\tau_{\geq 0}M)^{hC_p} \\ \downarrow & & \downarrow \\ M & \xrightarrow{F} & M^{hC_p} \end{array}$$

By university of $\tau_{\geq 0}(-)$, the composition is identified with the C_p-norm factored through the connective part, proving the assertion (2). The assertion (3) follows from the fact that the discrete topological Cartier module is equivalent to classical Cartier module. (4) follows immediately from construction. □

2.2. $\mathrm{TR}(-)$ **and cofiber of** V **adjunction.** Let M be a topological Cartier module, we consider the cofiber

$$M/V := \mathrm{cofib}\big(M_{hC_p} \xrightarrow{V} M\big)$$

in the category Sp^{BS^1}. The commutative square

$$\begin{array}{ccc} M_{hC_p} & \xrightarrow{\mathrm{id}} & M_{hC_p} \\ {\scriptstyle V}\downarrow & & \downarrow{\scriptstyle \mathrm{Nm}_{C_p}} \\ M & \xrightarrow[F]{} & M^{hC_p} \end{array}$$

induces a map $M/V \to M^{tC_p}$ by taking the cofiber. The composition $M/V \to M^{tC_p} \to (M/V)^{tC_p}$ makes M/V into a p-typical cyclotomic spectrum. This gives a functor

$$(-)/V \colon \mathsf{TCart}_p \to \mathsf{CycSp}_p$$

Lemma 2.11. *The functor* $(-)/V \colon \mathsf{TCart}_p \to \mathsf{CycSp}_p$ *is right t-exact.*

Proof. If $M \in (\mathsf{TCart}_p)_{\geq 0}$, then M_{hC_p} is connective, meaning the cofiber M/V is also connective. □

The functor $(-)/V$ preserves colimits. The adjoint functor theorem implies that $(-)/V$ admits a right adjoint. Our next goal is to identify this right adjoint.

Construction 2.12. Let $X \in \mathsf{CycSp}_p$. We define a spectrum $\mathrm{TR}^{n+1}(X)$ as

$$\mathrm{TR}^{n+1}(X) := \Big(X^{hC_{p^n}} \times_{(X^{tC_p})^{hC_{p^{n-1}}}} X^{hC_{p^{n-1}}} \times \cdots \times_{(X^{tC_p})^{hC_p}} X^{hC_p} \times_{X^{tC_p}} X\Big)$$

We define $\mathrm{TR}(X) := \lim_n \mathrm{TR}^{n+1}(X)$, which is also an S^1-equivariant spectrum. We have an S^1-equivariant equivalence

$$\Phi \colon \mathrm{TR}(X) \simeq X \times_{X^{tC_p}} \mathrm{TR}(X)^{hC_p}$$

where the arrow to the right is the Tate-value Frobenius and the arrow to the left is the composition of canonical map with the projection. We define the Frobenius as

$$F \colon \mathrm{TR}(X) \underset{\simeq}{\xrightarrow{\Phi}} X \times_{X^{tC_p}} \mathrm{TR}(X)^{hC_p} \xrightarrow{\mathrm{pr}_2} \mathrm{TR}(X)^{hC_p}$$

and the Verschiebung as

$$V\colon \mathrm{TR}(X)_{hC_p} \xrightarrow{(0,\mathrm{Nm})} X \times_{X^{tC_p}} \mathrm{TR}(X)^{hC_p} \xrightarrow[\simeq]{\Phi^{-1}} \mathrm{TR}(X)$$

It is clear that we have $F \circ V \simeq \mathrm{Nm}$. This endows $\mathrm{TR}(X)$ with a structure of topological Cartier module. We obtain a functor $\mathrm{TR}(-)\colon \mathsf{CycSp}_p \to \mathsf{TCart}_p$.

Finally, the composition $\mathrm{TR}(X)_{hC_p} \xrightarrow{V} \mathrm{TR}(X) \xrightarrow{\pi} X$ is null by construction. This induces a map $\mathrm{TR}(X)/V \to X$. This map can be refined to a map of cyclotomic spectra, which lifts to a nature transformation $\mathrm{TR}(-)/V \to \mathrm{id}\colon \mathsf{CycSp}_p \to \mathsf{CycSp}_p$.

Lemma 2.13. *Suppose X is connective, then $\mathrm{TR}(X)$ is also connective.*

Proof. Suppose that X is bounded below, by [NS18, Lemma II.4.1], there is an equivalence $X^{tC_{p^n}} \to (X^{tC_p})^{hC_{p^{n-1}}}$. In particular, we have an equivalence

$$\mathrm{TR}^{n+1}(X) := \left(X^{hC_{p^n}} \times_{X^{tC_{p^n}}} X^{hC_{p^{n-1}}} \times \cdots \times_{X^{tC_p}} X\right)$$

and we have a fiber sequence

$$X_{hC_p} \to \mathrm{TR}^{n+1}(X) \xrightarrow{R} \mathrm{TR}^n(R)$$

Note that $\mathrm{TR}^1(X) = X$. We can do induction. Computing along the fiber sequence shows that $\mathrm{TR}^{n+1}(X)$ is connective and on the zeroth homotopy group. So the $\lim^1$-term of coconnective homotopy groups vanishes after taking the limit, which shows that $\mathrm{TR}(X)$ is connective. □

Lemma 2.14 (Krause-Nikolaus)**.** *The projection $\mathrm{TR}(X) \to X$ exhibit $\mathrm{TR}(X)$ as the cofree p-cyclotomic spectrum with Frobenius lift over X. That is, $\mathrm{TR}(-)\colon \mathsf{CycSp}_p \to \mathsf{CycSp}_p^{\mathrm{Fr}}$ is right adjoint to the forgetful functor.*

Proof. We need to check the mapping space. The mapping space $\mathrm{Map}_{\mathsf{CycSp}_p^{\mathrm{Fr}}}(Y, \mathrm{TR}(X))$ in $\mathsf{CycSp}_p^{\mathrm{Fr}}$ is given by the equalizer:

$$\mathrm{Eq}\big((\psi_p)_*, \psi_p^*(-)^{hC_p}\colon \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, \mathrm{TR}(X)) \rightrightarrows \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, \mathrm{TR}(X)^{hC_p})\big)$$

Writing $\mathrm{TR}(-)$ as the iterated pullback, we found that this equalizer is equivalent to the total equalizer of the diagram

$$\begin{array}{ccc} \prod_n \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, X^{hC_{p^n}}) & \rightrightarrows & \prod_n \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, X^{hC_{p^{n+1}}}) \\ \downarrow\downarrow & & \downarrow\downarrow \\ \prod_n \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, (X^{tC_p})^{hC_{p^n}}) & \rightrightarrows & \prod_n \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, (X^{tC_p})^{hC_{p^{n+1}}}) \end{array}$$

The horizontal arrows are given by forgetting the first coordinate and $\psi_p^*(-)^{hC_p}$. The vertical arrows are Tate-valued Frobenius and the canonical map. Passing to the equalizer of horizontal maps, it becomes the equalizer

$$\mathrm{Eq}\big(\mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, X) \rightrightarrows \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(Y, X^{tC_p})\big)$$

which exactly computes the mapping space in the category of p-typical cyclotomic spectra. □

Proposition 2.15. *The functor* $\mathrm{TR}(-)\colon \mathsf{CycSp}_p \to \mathsf{TCart}_p$ *is right adjoint to* $(-)/V$*, with the counit given by the natural transformation* $\mathrm{TR}(-)/V \to \mathsf{TCart}_p$ *defined above.*

Proof. We consider the fiber sequence in Lemma 2.8:

$$\mathrm{Map}_{\mathsf{TCart}_p}(M, \mathrm{TR}(X)) \longrightarrow \mathrm{Map}_{\mathsf{CycSp}_p^{\mathrm{Fr}}}(M, \mathrm{TR}(X)) \longrightarrow \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(M_{hC_p}, \mathrm{fib}(F))$$

By Lemma 2.14, the middle term can be replaced by $\mathrm{Map}_{\mathsf{CycSp}_p}(M, X)$ and the right term can be replaced by $\mathrm{Map}_{\mathsf{CycSp}_p}(M_{hC_p}, X)$. The connecting map is given by precomposition with $V\colon M_{hC_p} \to M$ so we identify $\mathrm{Map}_{\mathsf{TCart}_p}(M, \mathrm{TR}(X))$ with $\mathrm{Map}_{\mathsf{CycSp}_p}(M/V, X)$. □

Remark 2.16. Let X be a bounded below cyclotomic spectrum, which means the underlying spectrum of X is bounded below, then the counit map $\mathrm{TR}(X)/V \to X$ is an equivalence. Indeed, consider the following diagram between null-homotopy columns:

$$\begin{array}{ccccc} \mathrm{TR}(X)_{hC_p} & \xrightarrow{V} & \mathrm{TR}(X) & \xrightarrow{\pi} & X \\ {\scriptstyle \mathrm{id}}\downarrow & & {\scriptstyle F}\downarrow & & \downarrow{\scriptstyle \varphi} \\ \mathrm{TR}(X)_{hC_p} & \xrightarrow{\mathrm{Nm}} & \mathrm{TR}(X)^{hC_p} & \longrightarrow & X^{tC_p} \end{array}$$

Then $\mathrm{TR}(X)/V \to X$ is an equivalence if and only if $\mathrm{TR}(X)^{tC_p} \simeq X^{tC_p}$. Note that we have a cofiber sequence

$$\mathrm{TR}^n(X)_{hC_p} \longrightarrow \mathrm{TR}^{n+1}(X) \longrightarrow X$$

Since X is bounded below, [NS18, Lemma II.4.1] implies that $\mathrm{TR}^n(X)$ is a iterated pullback along $X^{hC_{p^k}} \to X^{tC_{p^k}}$, which means that $\mathrm{TR}^n(X)$ and $\mathrm{TR}(X)$ are uniformly bounded below. Applying the Tate orbit lemma shows that we have $\mathrm{TR}^{n+1}(X)^{tC_p} \simeq X^{tC_p}$. Since for uniformly bounded below objects the Tate construction commutes with sequential limits, then we obtain that $\mathrm{TR}(X)^{tC_p} \simeq X^{tC_p}$, as desired.

Our next goal is to study the image of $\mathrm{TR}(-)$, or at least its image when restricting to bounded below objects. We will need some notations.

Definition 2.17. We say that a bounded below object $M \in \mathsf{TCart}_p^-$ is *V-complete* if the limit of the tower

$$\cdots \to M_{hC_{p^2}} \xrightarrow{V_{hC_p}} M_{hC_p} \xrightarrow{V} M$$

vanishes. We let $\mathsf{TCart}_p^{\wedge,-} \subset \mathsf{TCart}_p^-$ denote the full subcategory of V-complete objects. We say $M \to \hat{M}$ is V-completion if it induces an equivalence $\mathrm{Map}_{\mathsf{TCart}_p}(\hat{M}, N) \simeq \mathrm{Map}_{\mathsf{TCart}_p}(M, N)$ for every V-complete object N.

Proposition 2.18. *Let* $M \in \mathsf{TCart}_p^-$*, then:*

(1) $\mathrm{TR}(M/V) \simeq \lim_n M/V^n$*. Here* V^n *is the composition* $V_{hC_{p^{n-1}}} \circ \cdots \circ V$*.*
(2) $\mathrm{TR}(M/V)$ *is the* V*-completion of* M*.*

Proof. The assertion (1) follows from writing $\mathrm{TR}(M/V)$ as a sequential limit and a termwise comparison. For the second assertion, first note that for any bounded below X, we can

always replace $V_{hC_{p^{n-1}}}\colon \mathrm{TR}(X)_{hC_{p^n}} \to \mathrm{TR}(X)_{hC_{p^{n-1}}}$ by termwise maps

$$\cdots \times (X^{hC_p})_{hC_p^n} \times_{((X^{tC_p})^{hC_p})_{hC_p^n}} X_{hC_{p^n}} \times 0 \to$$
$$\cdots \times (X^{hC_{p^2}})_{hC_p^{n-1}} \times_{((X^{tC_p})^{hC_{p^2}})_{hC_p^{n-1}}} (X^{hC_p})_{hC_{p^{n-1}}} \times X_{hC_p^{n-1}}$$

with vanishing limits. This implies that $\mathrm{TR}(X)$ is always V-complete. For every V-complete object N we have

$$\begin{aligned}\mathrm{Map}_{\mathsf{TCart}_p}(\mathrm{TR}(M/V), N) \simeq& \mathrm{Map}_{\mathsf{TCart}_p}(\mathrm{TR}(M/V), \mathrm{TR}(N/V))\\ \simeq& \mathrm{Map}_{\mathsf{CycSp}_p}(M/V, N/V)\\ \simeq& \mathrm{Map}_{\mathsf{TCart}_p}(M, N)\end{aligned}$$

The first equivalence is because $N \simeq \lim N/V^n \simeq \mathrm{TR}(N/V)$ by completeness and (1), the second assertion is by adjunction and Remark 2.16, the last equivalence is by adjunction and completeness. This proves the assertion (2). □

We can conclude the above results as follows:

Theorem 2.19. *The right adjoint* $\mathrm{TR}(-)\colon \mathsf{CycSp}_p^- \to \mathsf{TCart}_p^-$ *is fully faithful, t-exact with the essential image identified with all V-complete topological Cartier modules.*

Proof. Fully faithfulness follows from the above proof. t-exactness follows from fully faithfulness and adjunction, and the essential image follows from the above proposition. □

2.3. **The heart of the cyclotomic t-structure.** Now we are going to identify the heart of the cyclotomic t-structure. It follows from Theorem 2.19 that the heart $\mathsf{CycSp}_p^\heartsuit$ is the subcategory of $\mathsf{TCart}_p^\heartsuit$ consisted of V-complete objects. We need to give a more intrinsic description.

Definition 2.20. Let $M \in \mathrm{Cart}_p$ be a classical p-typical Cartier module. We say that M is V-complete if it is V-complete in the derived sense. That is, the limit $\lim(\cdots \to M \xrightarrow{V} M \xrightarrow{V} M)$ vanishes in the derived category $\mathcal{D}(\mathbb{Z})$. We denote the full subcategory of V-complete objects by $\mathrm{Cart}_p^\wedge$.

Remark 2.21. Let M be a topological Cartier module that concentrated in a single degree, then two notations of V-completeness coincide. To see this, first suppose M is V-complete in the classical sense, then the limit $\lim(\cdots \to M_{hC_p^2} \xrightarrow{V_{hC_p}} M_{hC_p} \xrightarrow{V} M)$ can be written as a diagonal limit of the diagram:

$$\begin{array}{ccccc}
\vdots & & \vdots & & \vdots \\
\downarrow & & \downarrow & & \downarrow \\
\cdots \longrightarrow M \otimes BC_{p^2} & \xrightarrow{\mathrm{id}\otimes p} & M \otimes BC_p & \xrightarrow{\mathrm{id}\otimes p} & M \\
\downarrow{\scriptstyle V\otimes\mathrm{id}} & & \downarrow{\scriptstyle V\otimes\mathrm{id}} & & \downarrow{\scriptstyle V} \\
\cdots \longrightarrow M \otimes BC_{p^2} & \xrightarrow{\mathrm{id}\otimes p} & M \otimes BC_p & \xrightarrow{\mathrm{id}\otimes p} & M
\end{array}$$

The rightmost column already has vanishing limit, and all column has vanishing limit since M is bounded below so the sequential colimit commutes with homotopy fixed points. Conversely, suppose that X is topological V-complete which means the V-tower must admits vanishing

limits. We can take the induced limit on homotopy groups, whose lim and $\lim^1$ vanish by degree reason. This implies that X is V-complete in the classical sense.

Remark 2.22. A bounded below topological Cartier module is V-complete if and only if all its homotopy groups are derived V-complete. Indeed, suppose M is V-complete, then its zeroth homotopy group is derived V-complete by the above remark. We can apply $\tau_{\geq 1}M \to M \to \pi_0 M$ and do induction. Conversely, since the V-completion commutes with sequential limits, we can write M as a limit of its Postnikov tower and assume M is bounded. Using the extension, we can assume further that M concentrates in a single degree and apply the above remark.

Theorem 2.23. *Consider the fully faithful embedding induced by* $\mathrm{TR}(-)\colon \mathsf{CycSp}_p^{\heartsuit} \subset \mathsf{TCart}_p^{\heartsuit}$. *Let* $M \in \mathsf{TCart}_p^{\heartsuit}$. *The followings are equivalent:*

(1) M sits in $\mathsf{CycSp}_p^{\heartsuit}$.
(2) $M \to \mathrm{TR}(M/V)$ is an equivalence.
(3) M is a V-complete Cartier module.

Proof. Suppose that $M \in \mathsf{CycSp}_p^{\heartsuit}$, then M is V-complete topological Cartier module. By Proposition 2.18, this implies $M \to \mathrm{TR}(M/V)$ is an equivalence, proving (1) $\Rightarrow$ (2). (2) $\Rightarrow$ (3) is clear from Remark 2.21. Finally, (3) $\Rightarrow$ (1) follows from the description of the essential image by Theorem 2.19. □

Example 2.24. The category $\mathsf{CycSp}_p^{\heartsuit}$ is not compatible with filtered colimits. For every Abelian group, we can consider it as a Cartier module with $V = p$ and $F = \mathrm{id}$. In this case, V-completion is equivalent to derived p-completion. We consider

$$\mathrm{colim}_n \mathbb{Z}/p^n \simeq \mathbb{Q}_p/\mathbb{Z}_p$$

where each $\mathbb{Z}/p^n$ is p-complete. While this is not true for the right hand side since the derived p-completion of $\mathbb{Q}_p/\mathbb{Z}_p$ is $\mathbb{Z}_p[1]$.

2.4. **Symmetric monoidal structures.** By formal argument, since $(\mathsf{CycSp}_p)_{\geq 0}$ is compatible with the symmetric monoidal structure, the localization $\pi_0\colon (\mathsf{CycSp}_p)_{\geq 0} \to \mathsf{CycSp}_p^{\heartsuit}$ is symmetric monoidal, $\mathsf{CycSp}_p^{\heartsuit}$ inherits a symmetric monoidal structure. Now we are going to study the symmetric monoidal structure over $\mathsf{CycSp}_p^{\heartsuit}$. We will begin with a symmetric monoidal structure over $\mathsf{TCart}_p^{\wedge,-}$.

Definition 2.25. Let M and N be objects in $\mathsf{TCart}_p^{\wedge,-}$. We denote

$$M \hat{\boxtimes} N := \mathrm{TR}(M/V \otimes N/V)$$

We need to give a more concrete description of this symmetric monoidal structure. To do this, let us consider the forgetful functor $\mathsf{TCart}_p \to \mathsf{CycSp}_p^{\mathrm{Fr}}$.

Lemma 2.26. *The forgetful functor* $\mathsf{TCart}_p \to \mathsf{CycSp}_p^{\mathrm{Fr}}$ *is monadic. The underlying spectrum of the left adjoint is given by* $M[V] := \bigoplus_{n\geq 0} M_{hC_{p^n}}$.

Proof. The monadicity is clear since this forgetful functor preserves limits and colimits. We first construct the desired topological Cartier module structure over $M[V]$. We define

$$V\colon M[V]_{hC_p} \simeq \bigoplus_{n\geq 1} M_{hC_{p^n}} \hookrightarrow \bigoplus_{n\geq 0} M_{hC_{p^n}} \simeq M[V]$$

and

$$F\colon M[V] \to \bigoplus_{n\geq 0}(M_{hC_{p^n}})^{hC_p} \to \Big(\bigoplus_{n\geq 0} M_{hC_{p^n}}\Big)^{hC_p} \simeq M[V]^{hC_p}$$

where the first map is the direct sum of the Frobenius with the C_p-norms and the second map is canonical comparison. Since the norm map of $M[V]$ factors through

$$\bigoplus_{n\geq 1} M_{hC_{p^n}} \xrightarrow{\oplus \mathrm{Nm}_{C_p}} \bigoplus_{n\geq 0}(M_{hC_{p^{n-1}}})^{hC_p} \to \Big(\bigoplus_{n\geq 0} M_{hC_{p^{n-1}}}\Big)^{hC_p}$$

we have $F\circ V \simeq \mathrm{Nm}_{C_p}$. This gives the desired topological Cartier module over $M[V]$. It remains to show that $M[V]$ gives the desired left adjoint. By Remark 2.9, we have a cofiber sequence

$$\mathrm{Map}_{\mathsf{TCart}_p}(M[V],N) \longrightarrow \mathrm{Map}_{\mathrm{Alg}_{(-)_{hC_p}}}(M[V],N) \longrightarrow \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(\mathrm{cofib}(V), N^{hC_p})$$

where $\mathrm{cofib}(V)$ is identified with M and $\mathrm{Map}_{\mathrm{Alg}_{(-)_{hC_p}}}(M[V],N) \simeq \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(M,N)$ since $M[V]$ is the free $(-)_{hC_p}$-algebra. Therefore the cofiber sequence is identified with

$$\mathrm{Map}_{\mathsf{TCart}_p}(M[V],N) \longrightarrow \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(M,N) \longrightarrow \mathrm{Map}_{\mathsf{Sp}^{BS^1}}(M,N^{hC_p})$$

where the fiber of the second map is exactly the mapping space in $\mathsf{CycSp}_p^{\mathrm{Fr}}$, proving the claim. □

We consider the adjunction restricting to the bounded below part

$$(\mathsf{CycSp}_p^{\mathrm{Fr}})^- \rightleftharpoons \mathsf{TCart}_p^{\wedge,-}$$

The left adjoint sends M to the V-completion of $M[V]$, which is $\prod_{n\geq 0} M_{hC_{p^n}}$. Note that the mod V reduction of $M[V]$ is M, which implies that we also have the left adjoint sending M to $\mathrm{TR}(M)$. We will denote this left adjoint by $M[\![V]\!]$, note that $(-)[\![V]\!]\colon (\mathsf{CycSp}_p^{\mathrm{Fr}})^- \to \mathsf{TCart}_p^{\wedge,-}$ is symmetric monoidal.

Corollary 2.27. *For $M,N\in \mathsf{TCart}_p^{\wedge,-}$, $M\hat{\boxtimes}N \in \mathsf{TCart}_p^{\wedge,-}$ is identified with the total cofiber of*

$$\begin{array}{ccc} (M_{hC_p}\otimes N_{hC_p})[\![V]\!] & \longrightarrow & (M\otimes N_{hC_p})[\![V]\!] \\ \downarrow & & \downarrow \\ (M_{hC_p}\otimes N)[\![V]\!] & \longrightarrow & (M\otimes N)[\![V]\!] \end{array}$$

Proof. Since $(-)[\![V]\!]$ is symmetric monoidal, the square is equivalent to

$$\begin{array}{ccc} M_{hC_p}[\![V]\!]\hat{\boxtimes}N_{hC_p}[\![V]\!] & \longrightarrow & M[\![V]\!]\hat{\boxtimes}N_{hC_p}[\![V]\!] \\ \downarrow & & \downarrow \\ M_{hC_p}[\![V]\!]\hat{\boxtimes}N[\![V]\!] & \longrightarrow & M[\![V]\!]\hat{\boxtimes}N[\![V]\!] \end{array}$$

The assertion follows from the cofiber sequence $M_{hC_p}[\![V]\!]_{hC_p} \to M[\![V]\!] \to M$ for every M. □

Warning 2.28. In the first diagram of Corollary 2.27, the right and bottom arrows are not coming from maps in $\mathsf{CycSp}_p^{\mathrm{Fr}}$. It therefore demands additional efforts to identify these maps.

Now we can pass to the homotopy group. For any connective M, note that we have $\pi_0(M[\![V]\!]) \simeq \pi_0(M)[\![V]\!]$ is the power series generated by V. Assume that M and N are connective V-complete p-typical topological Cartier modules, after applying π_0, the homotopy group $\pi_0(M\hat{\boxtimes}N)$ is identified with the total cofiber of

$$\begin{array}{ccc} (\pi_0(M)\otimes\pi_0(N))[\![V]\!] & \longrightarrow & (\pi_0(M)\otimes\pi_0(N))[\![V]\!] \\ \downarrow & & \downarrow \\ (\pi_0(M)\otimes\pi_0(N))[\![V]\!] & \longrightarrow & (\pi_0(M)\otimes\pi_0(N))[\![V]\!] \end{array}$$

Let us have a look at what are these arrows. The left and the above arrows are from the cyclotomic spectra maps $V\otimes \mathrm{id}\colon M_{hC_p}\otimes N_{hC_p}\to M\otimes N_{hC_p}$ and $\mathrm{id}\otimes V\colon M_{hC_p}\otimes N_{hC_p}\to M_{hC_p}\otimes N$, since both the source and the target have the trivial Frobenius. As a consequence, on homotopy groups they send $(m\otimes n)V^k$ to $(V_M m\otimes n)V^k$ and $(m\otimes V_N n)V^k$. For the right arrow, however, since it is $V\otimes\mathrm{id}$ after mod V, the image of $m\otimes n$ must be of the form $m\otimes V_N n+$ higher terms. Since this right arrow must be compatible with the Frobenius, we can determine this higher term as $(-F_M m\otimes n)V$, together with V-linearity one can write down every map. Therefore, one shows the following:

Corollary 2.29. *Let $M,N\in \mathrm{Cart}_p^\wedge\simeq \mathsf{CycSp}_p^\heartsuit$. Then their symmetric monoidal product $M\hat{\boxtimes}N\in\mathsf{CycSp}_p^\heartsuit$ is defined by*

$$M\hat{\boxtimes}N\cong M\otimes N[\![V]\!]/\sim$$

where the equivalence is generated additively by

$$(m\otimes V_N n)V^k\sim (F_M m\otimes n)V^{k+1}\quad (F_M m\otimes n)V^k\sim (m\otimes V_N n)V^{k+1}$$

for every m, n and k.

Remark 2.30. It is clear that $\mathbb{Z}_p$, viewing as a V-complete Cartier module by $V=p$ and $F=\mathrm{id}$, becomes the unit in $\mathsf{CycSp}_p^\heartsuit$.

Remark 2.31. Let R be any classical commutative ring. Then we have $\pi_0^{\mathrm{cyc}}(\mathrm{THH}(R))\simeq \pi_0(\mathrm{TR}(R,p))\simeq W_p(R)$. Therefore, we have:

$$\begin{aligned} W(R\otimes S)&\cong \pi_0^{\mathrm{cyc}}(\mathrm{THH}(R)\otimes_{\mathrm{THH}(\mathbb{Z})}\mathrm{THH}(S))\\ &\cong \pi_0^{\mathrm{cyc}}\mathrm{THH}(R)\hat{\boxtimes}_{\pi_0^{\mathrm{cyc}}\mathrm{THH}(\mathbb{Z})}\pi_0^{\mathrm{cyc}}\mathrm{THH}(S)\\ &\cong W(R)\hat{\boxtimes}W(S)\end{aligned}$$

This can also be given a purely algebraic proof.

3. Genuine equivariant explanation

There is a parallel story building from genuine equivariant homotopy theory. We are going to compare the genuine equivariant explanation of topological Cartier modules and $\mathrm{TR}(-)$.

Definition 3.1. The ∞-category of p-typical genuine topological Cartier modules is defined as

$$\mathsf{TCart}_p^{\mathrm{gen}}:=\mathrm{Fix}_{(-)^{C_p}}\mathbb{T}\mathsf{Sp}_p^{\mathrm{gen}}:=\mathrm{Eq}\big(\mathrm{id},(-)^{C_p}\colon \mathbb{T}\mathsf{Sp}_p^{\mathrm{gen}}\to\mathbb{T}\mathsf{Sp}_p^{\mathrm{gen}}\big)$$

That means a p-typical genuine topological Cartier module is a p-typical genuine S^1-spectrum M with an equivalence $M \simeq M^{C_p}$.

Recall that the category of p-typical genuine cyclotomic spectra is defined as

$$\mathsf{CycSp}_p^{\mathrm{gen}} := \mathrm{Fix}_{\Phi^{C_p}(-)}\mathbb{T}\mathsf{Sp}_p^{\mathrm{gen}} := \mathrm{Eq}\big(\mathrm{id}, \Phi^{C_p}(-)\colon \mathbb{T}\mathsf{Sp}_p^{\mathrm{gen}} \to \mathbb{T}\mathsf{Sp}_p^{\mathrm{gen}}\big)$$

By a formal argument, we have the following:

Proposition 3.2. *There is an adjunction*

$$L\colon \mathsf{TCart}_p^{\mathrm{gen}} \rightleftarrows \mathsf{CycSp}_p^{\mathrm{gen}} : \mathrm{TR}^{\mathrm{gen}}(-)$$

where we have

$$L(M) \simeq \mathrm{colim}\big(M \to \Phi^{C_p}M \to \Phi^{C_{p^2}}(M) \to \cdots\big)$$

and

$$\mathrm{TR}^{\mathrm{gen}}(-) \simeq \lim\big(\cdots \to X^{C_{p^2}} \to X^{C_p} \to X\big)$$

Here the connection map is induced by $X^{C_p} \to \Phi^{C_p}(X)$.

Proof. This is quite formal: suppose we have an ∞-category $\mathcal{C}$ and a natural transformation of endofunctors $F \to G$ over $\mathcal{C}$, one can show that the colimit and limit tower naturally gives an adjunction

$$L\colon \mathrm{Fix}_F(\mathcal{C}) \rightleftarrows \mathrm{Fix}_G(\mathcal{C}) : R$$

□

We will use the following lemma, which can be viewed as a genuine version of Tate-orbit lemma.

Lemma 3.3. *For every object $M \in \mathbb{T}\mathsf{Sp}_p^{\mathrm{gen}}$, the canonical map $\Phi^{C_p}(M^{C_p}) \to \Phi^{C_{p^2}}M$ is an equivalence.*

Corollary 3.4. *Let $M \in \mathsf{TCart}_p^{\mathrm{gen}}$. There is a natural equivalence*

$$(LM)^{C_{p^n}} \simeq M/V^{n+1}$$

where V^{n+1} denotes the map $M_{hC_{p^{n+1}}} \to M^{C_{p^{n+1}}} \simeq M$.

Proof. Since the categorical fixed points commute with colimits, we have

$$(L(M))^{C_{p^n}} \simeq \mathrm{colim}\big(M^{C_{p^n}} \to (\Phi^{C_p}M)^{C_{p^n}} \to (\Phi^{C_{p^2}}M)^{C_{p^n}} \to \cdots\big)$$

The connection maps by Lemma 3.3 are equivalence after the second term in the sequence. As a result, the colimit is equivalent to $(\Phi^{C_p}M)^{C_{p^n}}$. Consider the fiber sequence

$$M_{hC_{p^{n+1}}} \longrightarrow M^{C_{p^{n+1}}} \longrightarrow (\Phi_{C_p}M)^{C_{p^n}}$$

After identifying the middle term with M, the first arrow becomes V^{n+1}. This proves the assertion. □

Now we are going to identify two category of topological Cartier modules.

Proposition 3.5. *The canonical forgetful functor $\mathsf{TCart}_p^{\mathrm{gen}} \to \mathsf{TCart}_p$ is an equivalence of ∞-category.*

Proof. Let $\mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^n}}$ denote the ∞-category of genuine equivariant spectra for the family of subgroups of $C_{p^n} \subseteq S^1$. Note that $\mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^0}} \simeq \mathsf{Sp}^{BS^1}$. We define

$$(\mathsf{TCart}^{\mathrm{gen}}_p)_n := \mathrm{Eq}\big((-)^{C_p}, U_{n+1} \colon \mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^{n+1}}} \rightrightarrows \mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^n}}\big)$$

Here $(-)^{C_p}$ and U_{n+1} are fixed points and forgetful functors respectively. By construction, there is a canonical map $\mathrm{can} \colon (\mathsf{TCart}^{\mathrm{gen}}_p)_n \to (\mathsf{TCart}^{\mathrm{gen}}_p)_{n-1}$ induced by the commutative diagram

$$\begin{array}{ccc}
\mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^{n+1}}} & \overset{(-)^{C_p}}{\underset{U_{n+1}}{\rightrightarrows}} & \mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^n}} \\
{\scriptstyle U_{n+1}}\downarrow & & \downarrow{\scriptstyle U_n} \\
\mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^n}} & \overset{(-)^{C_p}}{\underset{U_n}{\rightrightarrows}} & \mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^{n-1}}}
\end{array}$$

The resulting commutative diagram

$$\begin{array}{ccc}
\mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^{n+1}}} & \xrightarrow{(-)^{C_p}} & \mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^n}} \\
{\scriptstyle U_{n+1}}\downarrow & & \downarrow{\scriptstyle U_n} \\
\mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^n}} & \xrightarrow{(-)^{C_p}} & \mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_{C_{p^{n-1}}}
\end{array}$$

is a pullback square, thus it follows that for $n \geq 1$ the canonical map induces an equivalence $(\mathsf{TCart}^{\mathrm{gen}}_p)_n \to (\mathsf{TCart}^{\mathrm{gen}}_p)_{n-1}$. Therefore, we have

$$\mathsf{TCart}^{\mathrm{gen}}_p \simeq \lim_n (\mathsf{TCart}^{\mathrm{gen}}_p)_n \simeq (\mathsf{TCart}^{\mathrm{gen}}_p)_0$$

It remains to identify the right hand side.

Let $\mathscr{F}$ be a family of subgroups of G, When G is S^1 and $\mathscr{F} = \{C_p\}$, then a genuine $\mathscr{F}$-equivariant spectrum is determined by the Tate square

$$\begin{array}{ccc}
X^{C_p} & \longrightarrow & \Phi^{C_p} X \\
\downarrow & \lrcorner & \downarrow \\
X^{hC_p} & \longrightarrow & X^{tC_p}
\end{array}$$

which is further equivalent to the composition 2-simplex

$$\begin{array}{ccc}
X_{hC_p} & \longrightarrow & X^{C_p} \\
& {\scriptstyle \mathrm{Nm}_{C_p}} \searrow & \downarrow \\
& & X^{hC_p}
\end{array}$$

Therefore, one obtains a pullback square

$$\begin{array}{ccc}
\mathbb{T}\mathsf{Sp}^{\mathrm{gen}}_p & \longrightarrow & \mathrm{Fun}(\Delta_2, \mathsf{Sp}^{BS^1}) \\
{\scriptstyle U}\downarrow & \lrcorner & \downarrow{\scriptstyle \partial_1} \\
\mathsf{Sp}^{BS^1} & \xrightarrow[\mathrm{Nm}_{C_p}]{} & \mathrm{Fun}(\Delta_1, \mathsf{Sp}^{BS^1})
\end{array}$$

By definition, we have

$$\begin{aligned}(\mathsf{TCart}_p^{\mathrm{gen}})_0 &\simeq \mathrm{Eq}\Big((-)^{C_p}, U\colon \mathbb{T}\mathsf{Sp}_{C_p}^{\mathrm{gen}} \rightrightarrows \mathsf{Sp}^{BS^1}\Big)\\ &\simeq \mathrm{Eq}\Big(\mathrm{ev}_1\circ \mathrm{pr}_2, \mathrm{pr}_1\colon \mathsf{Sp}^{BS^1} \times_{\mathrm{Fun}(\Delta_1,\mathsf{Sp}^{BS^1})} \mathrm{Fun}(\Delta_2,\mathsf{Sp}^{BS^1}) \rightrightarrows \mathsf{Sp}^{BS^1}\Big)\\ &\simeq \mathsf{Sp}^{BS^1} \times_{\mathrm{Fun}(\Delta_1,\mathsf{Sp}^{BS^1})\times \mathsf{Sp}^{BS^1}} \mathrm{Fun}(\Delta_2,\mathsf{Sp}^{BS^1})\\ &\simeq \mathsf{TCart}_p\end{aligned}$$

□

Remark 3.6. The pullback square

$$\begin{array}{ccc}\mathbb{T}\mathsf{Sp}_{C_{p^{n+1}}}^{\mathrm{gen}} & \xrightarrow{(-)^{C_p}} & \mathbb{T}\mathsf{Sp}_{C_{p^n}}^{\mathrm{gen}}\\ {\scriptstyle U_{n+1}}\downarrow & & \downarrow{\scriptstyle U_n}\\ \mathbb{T}\mathsf{Sp}_{C_{p^n}}^{\mathrm{gen}} & \xrightarrow{(-)^{C_p}} & \mathbb{T}\mathsf{Sp}_{C_{p^{n-1}}}^{\mathrm{gen}}\end{array}$$

can be seen by interpreting $T\mathsf{Sp}_{C_{p^n}}^{\mathrm{gen}}$ as $\mathrm{Fun}(\mathrm{Span}(C_{p^n}), \mathsf{Sp}^{BS^1})$. This pullback square follows from the pushout diagram

$$\begin{array}{ccc}\mathrm{Span}(C_{p^{n-1}}) & \xrightarrow{C_p\ltimes(-)} & \mathrm{Span}(C_{p^n})\\ {\scriptstyle i}\downarrow & \ulcorner & \downarrow{\scriptstyle i}\\ \mathrm{Span}(C_{p^n}) & \xrightarrow[C_p\ltimes(-)]{} & \mathrm{Span}(C_{p^{n+1}})\end{array}$$

where i is the inclusion, corresponding to the forgetful functor and $C_p \ltimes (-)$ is the semi-product, corresponding to the categorical fixed points.

Remark 3.7. Through the equivalence $\mathsf{TCart}_p \simeq \mathsf{TCart}_p^{\mathrm{gen}}$, a p-typical topological Cartier module determines a genuine C_p-spectrum by assigning $X^{C_p} \simeq X$, therefore it is naturally a p-typical genuine cyclotomic spectra.

Now we can combine everything together to obtain the conclusion.

Theorem 3.8. *For any genuine topological Cartier module M, let $V\colon M_{hC_p} \to M^{C_p} \simeq M$ denote the obvious composition. There is an adjunction*

$$L\colon \mathsf{TCart}_p \rightleftarrows \mathsf{CycSp}_p^{\mathrm{gen}} : \mathrm{TR}^{\mathrm{gen}}(-)$$

such that

1. *The left adjoint satisfies $L(M)^{C_{p^n}} \simeq M/V^n$.*
2. *When restricting bounded below objects, this restricts to an equivalence connecting everything*

$$\mathsf{TCart}_p^{\wedge,-} \simeq \mathsf{CycSp}_p^{-} \simeq \mathsf{CycSp}_p^{\mathrm{gen},-}$$

Proof. This is given by combining Proposition 3.2, Corollary 3.4 and Proposition 3.5. □

Remark 3.9. It seems to have much more chance to expect fully faithfulness of the right adjoint in Proposition 3.2 on full objects, not merely bounded below objects. This is still

false in genuine settings. To see this, note that the counit map $\mathrm{TR}(X)/V \to X$ comes from the canonical comparison

$$\begin{aligned}\mathrm{TR}(X)/V &\simeq \mathrm{cofib}\big(\mathrm{TR}(X)_{hC_p} \xrightarrow{V} \mathrm{TR}(X)\big)\\ &\simeq \mathrm{cofib}\big((\lim X^{C_{p^n}})_{hC_p} \xrightarrow{V} (\lim X^{C_{p^n}})\big)\\ &\to \mathrm{cofib}\big(\lim (X^{C_{p^n}})_{hC_p} \xrightarrow{\lim V} \lim (X^{C_{p^n}})\big)\\ &\simeq \lim \mathrm{cofib}\big((X^{C_{p^n}})_{hC_p} \to X^{C_{p^n}}\big)\\ &\simeq \lim X \simeq X\end{aligned}$$

It can be computed that this comparison map is not an equivalence for general unbounded objects, say for instance $\mathrm{KU}^{\mathrm{triv}}$.

4. Applications

We now explain some applications of topological Cartier modules and the cyclotomic t-structure.

4.1. **Structure of the homotopy groups.** We show that for a topological Cartier module, the homotopy ring of the underlying spectra is equipped with more structure. Especially, it is a Dieudonné module.

Let $\mathbb{S}[S^1]$ denote the ∞-group ∞-ring generated by S^1. In particular, $\mathbb{S}[S^1]$ naturally splits, so we have $\pi_*(\mathbb{S}[S^1]) \simeq \pi_*(\mathbb{S})[d]/(d^2 - \eta d)$. Here η is the Hopf element and d is the Connes operator in degree 1. If M is a spectrum with S^1 action, then M becomes an $\mathbb{S}[S^1]$-module so it is equipped with a Connes action $d\colon \pi_*(M) \to \pi_{*+1}(M)$.

Proposition 4.1. *Let M be a topological Cartier module. Then the operators V, F, d over M satisfies the following relations:*

$$Vd = pdV, \quad dF = pFd, \quad FdV = \begin{cases} d & \text{if } p > 2 \\ d + \eta & \text{if } p = 2 \end{cases}$$

Proof. The modulo p equivalence induces a map over the group ring $\mathbb{S}[S^1/C_p] \to \mathbb{S}[S^1]$, which sends d to pd. Since V is S^1 equivariant, so V preserves the $\mathbb{S}[S^1]$-module structure, so on homotopy groups this gives $Vd = pdV$. Similarly, for the Frobenius we have $dF = pFd$. It remains to show the last relation. On homotopy groups the composition FdV can be written as the composition

$$\pi_*(M) \to \pi_*(M_{hC_p}) \xrightarrow{V_*} \pi_*(M) \xrightarrow{d} \pi_{*+1}(M) \xrightarrow{F_*} \pi_{*+1}(M^{hC_p}) \to \pi_{*+1}(M)$$

and it is further identified as the composition

$$\pi_*(M) \to \pi_*(M_{hC_p}) \xrightarrow{d'} \pi_{*+1}(M_{hC_p}) \xrightarrow{\mathrm{Nm}_{C_p}} \pi_{*+1}(M^{hC_p}) \to \pi_{i+1}(M)$$

By base-change, it suffices to show this map is identified with the given form for $M = \mathbb{S}[S^1]$. Writing $\mathbb{S}[S^1] \simeq \mathbb{S} \vee \Sigma\mathbb{S}$, the canonical map $\mathbb{S}[S^1] \to \mathbb{S}[S^1]_{hC_p} \simeq \mathbb{S}[S^1]$ is given by $\mathrm{id} \vee p$, the norm map is the identity and the map $\mathbb{S}[S^1]^{hC_p} \simeq \mathbb{S}[S^1] \to \mathbb{S}[S^1]$ is given by

$$\begin{pmatrix} p & (p-1)\eta \\ 0 & \mathrm{id} \end{pmatrix} : \mathbb{S} \vee \Sigma\mathbb{S} \to \mathbb{S} \vee \Sigma\mathbb{S}$$

Keeping track of the image of $1 \in \pi_*(\mathbb{S})$ and using the fact that η is 2-torsion, we find the composition is exactly d when $p \neq 2$ and $d + \eta$ when $p = 2$.

□

Remark 4.2. In [BLM18, Section 2], we have the notation of Dieudonné complexes. The above proposition shows that when M is a topological Cartier module, its homotopy becomes a Dieudonné complex with additional structures.

4.2. **Descent property.** We give a quick discussion of the descent property of $\mathrm{TR}(-)$.

Definition 4.3. A map of (classical) commutative rings $f\colon R \to S$ is called *weakly étale* if $R \to S$ and $S \otimes_R S \to S$ are both flat. We let $\mathrm{CRing}^{\mathrm{proet}}$ denote the site generated by faithfully weakly étale covers and refer to it as the *pro-étale* site

Proposition 4.4. $\mathrm{THH}(-)\colon \mathrm{CRing} \to \mathsf{CycSp}$ *is a hypercomplete pro-étale sheaf.*

Sketch of proof. We first show that $\mathrm{THH}(-)$ is an Sp-value pro-étale sheaf. We can always write $\mathrm{THH}(S) \simeq \lim_n \mathrm{THH}(S) \otimes_{\mathrm{THH}(\mathbb{Z})} \tau_{\leq n}\mathrm{THH}(\mathbb{Z})$. Since the limit of hypercomplete sheaves are again hypercomplete, it remains to show each term $\mathrm{THH}(-) \otimes_{\mathrm{THH}(\mathbb{Z})} \tau_{\leq n}\mathrm{THH}(\mathbb{Z})$ is hypercomplete. By further truncation, it remains to show $\mathrm{HH}(-)$ is hypercomplete. Using the HKR filtration, we reduce to prove the graded pieces $\Lambda^n \mathrm{L}_{(-)/Z}$ is hypercomplete. This is clear since it is quasi-coherent.

For CycSp-value, it suffices to show that for any pro-étale hyper-cover $R \to S$, we have an equivalence

$$\mathrm{THH}(R)^{tC_p} \simeq \lim_{\Delta} \mathrm{THH}(S^\bullet)^{tC_p}$$

Again, this reduces to $(\Lambda^n \mathrm{L}_{(-)/Z})_{hC_p}$ and follows from quasi-coherence. □

Corollary 4.5. *The functor* $\mathrm{TR}(-)\colon \mathrm{CRing} \to \mathsf{TCart}_p$ *is a hypercomplete pro-étale sheaf.*

Proof. This follows immediately from that $\mathrm{TR}(-)$ preserves limits. □

Remark 4.6. One can show that $\mathrm{TR}(-) \to \lim_n \tau_{\leq n}\mathrm{TR}(-)$ becomes an equivalence after Zariski sheafification. Let $L_{\mathrm{zar}}\pi_n^{\mathrm{cyc}}(-)$ denote the Zariski sheafification of $\pi_n^{\mathrm{cyc}}(-)$, which is a Zariski sheaf with value in Cart_p. Then the Whitehead tower for $\mathrm{TR}(-)$ gives a conditionally convergent spectral sequence

$$E_2^{s,t} \cong \mathrm{H}^{-s}(X, L_{\mathrm{zar}}\pi_t^{\mathrm{cyc}}\mathrm{THH}(-)) \Rightarrow L_{\mathrm{zar}}\pi_{t+s}^{\mathrm{cyc}}\mathrm{THH}(X)$$

Proposition 4.7. *Suppose that R is a perfect ring of character p, then* $\mathrm{THH}(R) \in \mathsf{CycSp}_p^{\heartsuit}$ *and the associated Cartier module is exactly $W(R)$.*

Proof. We need to show that $\mathrm{TR}(R) \simeq \mathrm{TR}_0(R) \simeq W(R)$. When R is a perfect ring, this is already proven by [Hes97]. In general, after apply $(-)/V$, we need to show that the composition $\mathrm{THH}(R) \to \pi_0^{\mathrm{cyc}}\mathrm{THH}(R) \simeq W(R)/V$ is an equivalence. When R is perfect, by the computation of [BMS19], both $\mathrm{THH}(R)$ and $W(R)/V$ are isomorphic to $k[b]$ with b in degree 2. We need to show that $\mathrm{THH}(R) \to W(R)/V$ sends b to b. This reduces to the case $R = \mathbb{F}_p$ and is already true. □

Remark 4.8. Let k be a perfect field. Let R be a smooth k-algebra. Then $\pi_*^{\mathrm{cyc}}(\mathrm{THH}(R)) \simeq \mathrm{TR}_*(R) \simeq \mathrm{W}\Omega_R^*$, the de-Rham Witt complex, as Dieudonné complex. Therefore the spectral sequence in Remark 4.6 becomes

$$E_2^{s,t} \cong \mathrm{H}^{-s}(X, \mathrm{W}\Omega_{(-)}^t) \Rightarrow L_{\mathrm{zar}}\pi_{t+s}^{\mathrm{cyc}}\mathrm{THH}(X)$$

References

[AN21] Benjamin Antieau and Thomas Nikolaus. "Cartier modules and cyclotomic spectra". In: *Journal of the American Mathematical Society* 34.1 (2021), pp. 1–78 (cit. on p. 1).

[Ant+22] Benjamin Antieau et al. "On the Beilinson fiber square". In: *Duke Mathematical Journal* 171.18 (2022), pp. 3707–3806 (cit. on p. 1).

[BLM18] Bhargav Bhatt, Jacob Lurie, and Akhil Mathew. "Revisiting the de Rham-Witt complex". In: *arXiv preprint arXiv:1805.05501* (2018) (cit. on p. 17).

[BMS19] Bhargav Bhatt, Matthew Morrow, and Peter Scholze. "Topological Hochschild homology and integral p-adic Hodge theory". In: *Publications mathématiques de l'IHÉS* 129.1 (2019), pp. 199–310 (cit. on p. 17).

[Hes97] Lars Hesselholt. "Witt vectors of non-commutative rings and topological cyclic homology". In: (1997) (cit. on p. 17).

[Lur17] Jacob Lurie. *Higher Algebra*. available online, 2017. URL: https://people.math.harvard.edu/~lurie/papers/HA.pdf (cit. on pp. 2, 3, 5).

[NS18] Thomas Nikolaus and Peter Scholze. "On topological cyclic homology". In: (2018) (cit. on pp. 2, 3, 7, 8).